

SwasthyaSetu

AI-Powered Care Orchestration — Revised Feature & Product Blueprint

“Don’t just send the patient. Make sure the care is ready — and stay with the journey until care is complete.”

Purpose

This revised blueprint preserves the existing SwasthyaSetu architecture while making four focused product changes: **Patient Voice-First**, **Patient Offline-First**, a **low-burden coordinated One-Trip experience**, and **Care Readiness as the WOW demo feature**.

1. Core Product Idea

SwasthyaSetu is an intelligent care-orchestration system for rural and underserved public healthcare. It determines what care a patient needs, checks whether that care is actually ready at a destination, routes the patient dynamically, tracks every required care action, and intervenes when the journey gets stuck.

2. The Complete Patient Care Loop

#	Stage	What SwasthyaSetu Does
1	Patient Need	Assess symptoms, vitals, history and existing reports.
2	AI Understanding	Create a concise patient summary from structured/unstructured information.
3	Care Bundle	Identify specialty, diagnostics, consultation, treatment and follow-up requirements.
4	CARE READINESS ■	WOW CHECK: verify capability, doctor/diagnostic availability, appointment slots and capacity before travel/referral.
5	Best Route	Rank feasible facilities/routes and reduce unnecessary travel.
6	Referral	Send structured referral with priority and required information.
7	Care Journey	Track acceptance → appointment → arrival → consultation → diagnostics/treatment.
8	Intervention	Detect delays, identify the reason, intervene and re-route/recover.
9	Follow-up	Track due actions and prioritize follow-ups.
10	Care Completed	Close only when the required care bundle is actually completed.

3. AI & Intelligence Layer

AI Component	Purpose	Prototype Approach
Patient Understanding	Extract symptoms, history, medicines and key information.	LLM/NLP + structured extraction
AI Patient Summary	Turn scattered assessment/report data into a concise care summary.	LLM with controlled output
Care Requirement Model	Generate the required Care Bundle.	LLM + clinical rules/knowledge base
Care Readiness Engine ■	Determine whether a facility can actually deliver the required care now.	Rules + scoring/ML + simulated live facility data for demo
Facility Recommendation	Rank suitable facilities.	Recommendation/ranking model
Route Optimization	Find the best feasible care route.	Graph + optimization
Referral Delay Prediction	Estimate risk that a journey will become delayed/stuck.	XGBoost / Random Forest
Intervention Engine	Suggest the next recovery action.	Rules + ranking + workflow logic
Document Extraction	Extract information from medical reports.	OCR + NLP
Translation & Voice ■	Patient speaks in a local language; system converts voice into structured information and journey actions.	Speech-to-text + translation + LLM
Setu Saathi	Answer journey-aware patient questions.	LLM + patient journey data
Network Intelligence	Detect recurring system-level bottlenecks.	Analytics + anomaly/pattern detection

4. High-Value Product Features

Care Readiness — WOW Feature ■

Before a patient travels, SwasthyaSetu checks the destination against the required Care Bundle: **specialty + doctor/slot + diagnostics + equipment/capability + capacity + appointment status**. The demo can show a facility changing from READY to NOT READY and the system automatically selecting a feasible alternative.

Smart Referral Recovery

When a referral fails or a destination becomes unavailable, identify the reason and find an alternative facility/route instead of ending the journey.

Coordinated One-Trip Care

Optimize for **one planned trip for the patient** while reducing manual coordination for the ASHA. The system pre-checks readiness, appointments, diagnostics, documents and route so the patient is not sent back-and-forth and the ASHA does not have to coordinate each step manually.

Care Journey Timeline

Show Assessment → Referral → Acceptance → Appointment → Arrival → Consultation → Diagnostics/Treatment → Follow-up → Completed.

Referral Stuck Explanation

A clear “Why is this stuck?” view showing the operational reason and available recovery action.

ASHA Priority Queue

Daily queue of urgent cases, stuck referrals, upcoming appointments and follow-ups.

Smart Document Checklist

Before travel, show required ID, referral, reports, prescriptions and other documents; flag missing items.

Patient Travel Assistance

Show destination, distance, appointment time, required documents and journey instructions.

Emergency Escalation

Flag potentially urgent assessments to authorized healthcare personnel for faster review; AI does not replace clinical decisions.

Family/Caregiver Access

Consent-based access to referral status, appointments, next actions and follow-up information.

5. Patient-First Voice Experience

VOICE-FIRST PATIENT FLOW

Patient speaks → local-language speech-to-text → translation/understanding → structured symptoms & needs → AI Patient Summary → Care Bundle → Care Readiness → next action

Voice-First Patient Input

The primary voice interaction is now designed for the **patient**, not the ASHA. A patient can describe symptoms or questions in a local language; the system converts the conversation into structured information that can be reviewed before it is used for care orchestration.

District Bottleneck Heatmap

Visualize recurring specialist, diagnostic, capacity and appointment bottlenecks.

Referral Analytics

Show completed, pending, delayed and rejected referrals plus reasons for failure.

6. Patient Offline-First Architecture

Offline support is now explicitly patient-facing. The Patient App should remain useful when connectivity is weak or unavailable, storing permitted journey information and actions locally and syncing them when internet returns.

Patient Offline	When Internet Returns
View previously synced journey, referral and appointment information	Refresh current server state
View saved care instructions / documents	Refresh or sync updated information
Record permitted offline actions / questions	Upload queued events/actions
Queue changes locally	Process sync queue with conflict-aware handling

Sync principle: treat offline actions as events/queued changes rather than blindly overwriting server rows. Use sync status and conflict-aware handling when the same record changes in multiple places.

Offline limitation: live facility availability, live appointment booking and real-time status cannot be guaranteed without connectivity.

7. User Modules

Module	Key Functions
Patient App	Voice-first input, offline journey access, journey timeline, referral status, appointments, records, Setu Saathi, notifications, travel/document assistance.
ASHA / ANM / PHC	Registration, assessment, AI summary, Care Bundle, referral creation, monitoring and intervention tasks.
Receiving Facility / Doctor	Referral inbox, patient clinical view, accept/reject/request information, consultation, treatment, diagnostics and follow-up.
Diagnostics	Required tests, facility availability, bundled investigations and completion tracking.
District/Admin	Network map, bottleneck detection, referral-failure analysis, route simulation and network insights.

8. Recommended MVP for a Strong Demo

MVP Feature	Demo Purpose
■ Care Readiness WOW	Show whether the destination is truly ready for the patient's complete Care Bundle — capability, doctor/slot, diagnostics and capacity.
AI Patient Summary + Care Bundle	Turn patient input into an understandable care plan for review.
Smart Facility / Route Recommendation	Use readiness and travel constraints to select a feasible destination/route.
Coordinated One-Trip Optimization	Reduce patient travel and reduce ASHA coordination effort by bundling planned services.
Referral Tracking + Visual Journey	Make the complete journey visible from referral to completion.
Referral Delay Detection + Recovery	Demonstrate intervention when the chosen facility or referral path breaks.
Patient Voice + Offline Mode	Demonstrate access in local language and low-connectivity conditions.
Setu Saathi	Answer journey/status questions using patient journey data.
District Bottleneck Dashboard	Use simulated/demo data to show system-level bottlenecks.

9. Suggested Demo Story — Make Care Readiness the WOW Moment

Step	Moment	What Judges See
1	Patient Voice	Patient describes the problem in a local language. The app converts voice into structured information.
2	AI Understanding	AI creates a concise patient summary and proposes a Care Bundle for review.
3	Care Readiness ■	System checks Facility A: required specialty, doctor/slot, diagnostics, equipment/capability and capacity. Show a clear READY / NOT READY status.
4	One-Trip Planning	System bundles feasible consultation + investigations and checks documents/appointment timing so the patient is not sent unnecessarily multiple times.
5	Route + Referral	System recommends the feasible route and sends the structured referral. Patient sees the journey timeline.
6	Break the Journey	Simulate Facility A becoming unavailable. The system detects the readiness change / referral risk.
7	Referral Recovery	System identifies the reason, finds a feasible alternative and updates the patient journey.
8	Patient Offline	Turn off connectivity: patient can still view saved journey information and queued permitted actions.
9	Follow-up	ASHA receives the follow-up task and the patient receives the next action.
10	Care Completed	Required care actions are marked complete; only then is the journey closed.

DEMO WOW MOMENT

“The patient is not sent because a hospital exists. The patient is sent because the required care is READY.”

Show the Care Readiness panel first, then deliberately change one facility condition and let SwasthyaSetu recover the journey.

10. Care Readiness — Detailed Demo Architecture

Care Readiness is the decision layer between “the patient needs care” and “send the patient there.” It should visibly explain **why a facility is ready or not ready**, rather than showing only a single score.

Readiness Check	Example Demo Signal	Result
Required specialty	Cardiology available	READY
Doctor / appointment	Consultation slot available today	READY
Diagnostics	Required test available on-site	READY
Equipment / capability	Required equipment/service available	READY
Capacity	Capacity threshold not exceeded	READY
Documents	Required documents complete	READY

Recommended UI for the WOW Screen

PATIENT NEED	CARE BUNDLE	FACILITY READINESS
Specialty: Cardiology Priority: Routine Patient: Rural/underserved	Consultation ECG Blood tests Follow-up	READY ✓ Doctor slot ✓ Diagnostics ✓ Capacity ✓ Documents ✓

Failure Simulation

For the demo, change one operational variable — for example, the diagnostic slot becomes unavailable. The readiness panel should immediately change to **NOT READY**, explain the missing requirement, and trigger facility/route re-evaluation. This makes the AI orchestration visible instead of presenting a static hospital list.

Coordinated One-Trip Care — Burden Reduction

Without coordination	With SwasthyaSetu
Patient travels → discovers missing service/slot → returns → repeats journey.	Readiness checked → appointment aligned → diagnostics bundled → documents checked → one planned trip where feasible.
ASHA manually calls/checks multiple facilities and coordinates steps.	System presents a coordinated feasible plan and exceptions; ASHA reviews instead of manually coordinating every step.

Design Principle

One-Trip is an optimization goal, not a promise. If the complete Care Bundle cannot safely or operationally be completed in one trip, SwasthyaSetu should explain why and choose the next feasible plan.

11. Technical Architecture

Layer	Technology / Approach
Frontend	React or Flutter
Backend	FastAPI / Node.js
AI Orchestration	LLM + NLP + OCR + ML prediction + recommendation/routing services
Database	PostgreSQL; PostGIS for geographic/network information
Patient Offline Store	SQLite/local storage with a synchronization queue
Security	Role-based access, consent, audit logs, access history and data minimization
Integration-ready	Architecture can be prepared for ABDM/ABHA, eSanjeevani, HMIS, laboratory, pharmacy and GIS integrations when actual APIs/data connections exist.

12. Core USP

SwasthyaSetu is not simply a telemedicine app, hospital finder, EHR or referral tracker. Its core proposition is to coordinate the care journey around what the patient actually needs, **verify whether that care is ready**, adapt when the route breaks, and continue monitoring until the required care is completed.

JUDGE-FRIENDLY ONE-LINER

“SwasthyaSetu uses AI to understand what care a patient needs, verify where that care is actually ready, select the best feasible route, predict where the journey may fail, and intervene until the care bundle is completed.”

13. Important Prototype Rules

- AI recommendations support healthcare workers; final clinical decisions and referral actions remain with authorized personnel.
- Do not present AI output as a diagnosis unless validated and clinically authorized.
- Live availability/capacity should only be claimed when connected to real operational data.
- Prototype metrics, capacity figures and outcome improvements must be clearly labelled as simulated/demo data unless backed by real data.
- Do not claim ABDM/ABHA, eSanjeevani, HMIS or other integrations unless an actual API/data connection exists.
- **Care Readiness demo note:** if the demo uses simulated facility status, label it clearly as simulated data while demonstrating the decision logic.

Revision focus: Patient Voice-First • Patient Offline-First • Low-Burden One-Trip Coordination • Care Readiness WOW Demo

Source basis: SwasthyaSetu — Consolidated Product Blueprint (uploaded project document).